

Python Comprehension – Practical Questions

Practice questions covering List, Dictionary, Set, and Nested Comprehensions

List Comprehension – Practical Questions

1. Even Numbers

Given a list of numbers, create a new list containing only the even numbers using list comprehension.

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

2. Squares

Create a list containing the squares of numbers from 1 to 20 using list comprehension.

3. Numbers Greater Than 50

Extract all numbers greater than 50 from the following list.

```
numbers = [12, 67, 34, 89, 45, 90, 23, 56, 78]
```

4. Convert Strings to Uppercase

Convert every string in the following list to uppercase.

```
names = ["aftab", "rahul", "neha", "aman", "priya"]
```

5. Length of Words

Create a list containing the length of each word.

```
words = ["Python", "Machine", "Learning", "AI", "Developer"]
```

6. Find Vowels

From a given string, create a list containing only the vowels.

```
text = "artificial intelligence"
```

7. Positive Numbers

Extract only positive numbers from the following list.

```
numbers = [-5, 10, -2, 8, 0, -7, 15, 20]
```

8. Divisible by 3 and 5

Generate all numbers between 1 and 100 that are divisible by both 3 and 5.

Dictionary Comprehension – Practical Questions

9. Number and Square

Create a dictionary where the key is a number from 1–10 and the value is its square.

```
# Expected: {1: 1, 2: 4, 3: 9, ...}
```

10. Student Marks

Create a new dictionary containing only students who scored 50 or more.

```
students = {"Aftab": 85, "Rahul": 45, "Neha": 92, "Aman": 38, "Priya": 76}
```

11. Word Length Dictionary

Create a dictionary where each word is the key and its length is the value.

```
words = ["Python", "AI", "Machine", "Learning"]
```

12. Character Frequency

Create a dictionary containing each character and its frequency using dictionary comprehension.

```
text = "programming"
```

Set Comprehension – Practical Questions

13. Unique Squares

Create a set containing the squares of all unique numbers.

```
numbers = [1, 2, 2, 3, 4, 4, 5, 5]
```

14. Unique Word Lengths

Create a set containing the unique lengths of the words.

```
words = ["AI", "Python", "ML", "Developer", "Data"]
```

15. Unique Vowels

Extract all unique vowels from a string using set comprehension.

```
text = "Artificial Intelligence and Machine Learning"
```

Advanced Practical Questions

16. Nested List Comprehension

Flatten the following nested list into a single list.

```
matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
```

17. Matrix Transformation

Create a new matrix where every element is multiplied by 2 using nested list comprehension.

18. Even/Odd Dictionary

Create a dictionary from 1 to 20 where each number is mapped to "Even" or "Odd".

19. Filter Students by Marks

Given a student dictionary, create three dictionaries: students scoring ≥ 80 , students scoring 50–79, and students scoring < 50 .

20. Word Filtering

Given a sentence, create a list containing only words whose length is greater than 4.

```
sentence = "Python is a powerful programming language for artificial intelligence"
```

Student Exercise: Try Questions 1–15 without using explicit for-loops. For Questions 16–20, nested comprehensions and conditional expressions are allowed.